

NBA Player Performance Analysis

AI305 - Introduction to Machine Learning

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Project Overview and Objectives

The main goal of this project is to use machine learning techniques to analyze NBA player statistics on a game-by-game basis. The analysis aims to extract insights into player performance, player archetypes, and game outcomes.

Objectives

- Predict player performance
- Identify player archetypes
- Analyze the relationship between player statistics and game outcomes
- (Optional) Perform temporal performance trend analysis

Problem Definition

The project focuses on analyzing NBA player statistics using three machine learning tasks:

Task 1 – Regression

- **Target variable:** Player Points Per Game (PTS)
- **Goal:** Predict how many points a player will score in a given game.

Task 2 – Clustering

- **Approach:** K-Means (primary) and Gaussian Mixture Models (GMM)
- **Goal:** Group players into archetypes such as scorers, defenders, or playmakers.

Task 3 – Classification

- **Target variable:** Game Outcome (Win or Loss)
- **Goal:** Predict whether a team wins based on aggregated player statistics.

Open-Ended Task Selected

- **Temporal Performance Analysis:** An optional task to analyze player performance trends across time using rolling averages.

Dataset Description

The dataset contains per-game NBA player statistics collected from a full season.

Each row represents one player's performance in a single game.

Dataset Summary

- **Rows:** (add after checking `df.shape`)

- **Columns:** (add after checking)
- **Source:** Provided dataset (Kaggle/Instructor)
- **Type of data:** Numerical features describing player scoring, shooting efficiency, defense, and playmaking.

Key Variables

- **PTS:** Points scored
 - **AST:** Assists
 - **TRB:** Total rebounds
 - **MP:** Minutes played
 - **FGA/3PA/FTA:** Shooting attempts
 - **FG%, 3P%, FT%:** Shooting efficiency percentages
 - **STL, BLK:** Defensive contributions
 - **TOV:** Turnovers
 - **Plus/Minus:** On-court team impact
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Preliminary Data Analysis and Cleaning

Dataset Statistics

Summary statistics (mean, median, standard deviation, minimum, maximum) were computed for:

- PTS (Points)
- AST (Assists)
- TRB (Total Rebounds)
- MP (Minutes Played)

These statistics reveal variations in player roles and productivity and help identify outliers and typical performance levels.

Distribution and Correlation Analysis

Distribution Plots

- Histograms → Identify skewness, normality, outliers
- Boxplots → Detect unusual performances
- Scatterplots (PTS vs MP, PTS vs FGA) → Reveal linear and non-linear relationships

Correlation Analysis

A correlation matrix and heatmap revealed:

- Strong correlation between MP and PTS
- High correlations among offensive stats (PTS, FGA, 3PA)
- High correlations among defensive stats (STL, BLK, TRB)
- Lower correlation between playmaking and scoring stats

This helps identify multicollinearity and guides model selection.

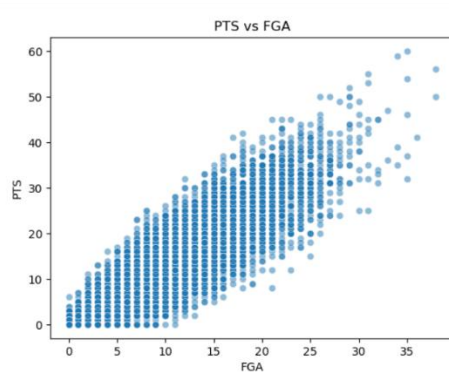
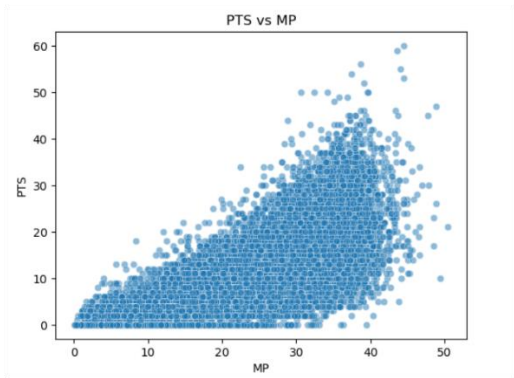
Handling Missing Values

- Percent-based stats with zero attempts (FG%, 3P%, FT%) → set to 0
- Games with MP = 0 → removed
- Corrupted or incomplete rows → cleaned

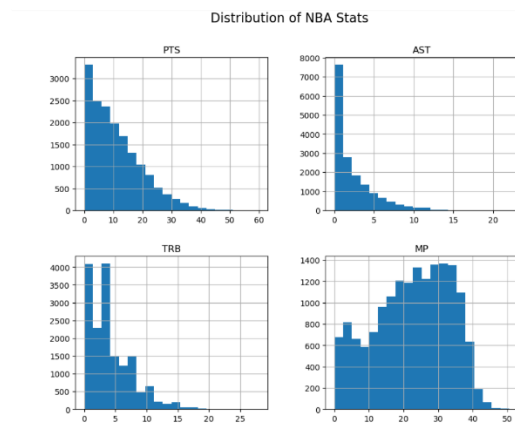
This ensures a consistent dataset for training.

Proposed Methodology and Tasks

The project includes three machine learning tasks: Regression, Clustering, and Classification, plus an optional temporal analysis task.



```
Missing values per column:
Player 0
Tm 0
Opp 0
Res 0
MP 0
FG 0
FGA 0
FG% 0
3P 0
3PA 0
3P% 0
FT 0
FTA 0
FT% 0
ORB 0
DRB 0
TRB 0
AST 0
STL 0
BLK 0
TOV 0
PF 0
PTS 0
GmSc 0
Data 0
dtype: int64
```



Engineered Features

Feature	Purpose	Formula
True Shooting % (TS%)	Scoring efficiency	$PTS / (2 \times (FGA + 0.44 \times FTA))$
Assist-to-Turnover Ratio (A/TOV)	Playmaking efficiency	$AST / (TOV + 1)$

Defensive Impact Score (DIS)	Defensive contribution	$(STL \times 2) + BLK +$ DRB
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These features improve representation of efficiency, playmaking, and defense.

Task 1: Regression – Predicting Player Points

Goal

Predict the number of points scored by a player in a game.

Models

- Linear Regression (Baseline; interpretable)
- Random Forest Regressor (handles non-linear patterns)
- Support Vector Regression (SVR) (effective for complex boundaries)

Evaluation Metrics

- RMSE
 - R^2 Score
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Task 2: Clustering – Identifying Player Styles

Goal

Group players based on their per-game average performance.

Models

- K-Means (required; effective after scaling)
- Gaussian Mixture Models (GMM) (allows overlapping clusters)

Clustering Steps

1. Compute per-game averages
 2. Standardized features
 3. Determine optimal k (Elbow + Silhouette Score)
 4. Use PCA for visualization
 5. Interpret clusters
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Task 3: Classification – Predicting Game Outcome

Goal

Predict whether a team wins or loses based on aggregated player statistics.

Models

- Logistic Regression (baseline; interpretable)
- Random Forest Classifier (handles non-linear boundaries)
- Support Vector Machine (SVM) (effective in high-dimensional spaces)

Evaluation Metrics

- Accuracy

- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC-AUC

At least one model will undergo hyperparameter tuning.

Bonus Task: Temporal Performance Analysis

Goal

Analyze performance changes over time.

Method

- Rolling averages (5-game or 10-game windows)
 - Identify slumps, improvements, or fatigue
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Tools and Libraries

Category	Tools
Programming Language	Python
Data Manipulation	pandas, numpy
Visualization	matplotlib, seaborn
Machine Learning	scikit-learn
Dimensionality Reduction	PCA, t-SNE

Contribution Breakdown

Team Member	Responsibilities
Juri	Data cleaning, descriptive stats, handling missing values, regression task
Nejoud	Distribution analysis, feature engineering, clustering task, bonus task
Shahad	Methodology writing, classification task, hyperparameter tuning
All Members	Final report and presentation preparation

Conclusion

This proposal presents a structured plan to analyze NBA player statistics using both supervised and unsupervised learning. Through regression, clustering, classification, and engineered features, the project aims to uncover meaningful insights about player performance, game outcomes, and trends across the season.